

Load Interconnection System Impact Study Report

For Queue Project Q12-0096

Shroud 100 kV

June 27, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0096. The Interconnection Customer proposes a 103.5 MW large load (data center) facility interconnecting at the Shroud 100 kV substation in the Indiana area. The request was received on January 16, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 79 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$86,000.

3. Network Impacts

3.1 Load Deliverability Analysis

EGE to GOLDEN GATE 100kV: 104.5% loading (154 MVA) following the outage of the DEKALB to JONES CHAPEL 500kV Line; DFAX = 5.3%.

The EGE to CUBA CITY 100kV is overloaded to 101.8% of its 146 MVA emergency rating for the loss of the DEKALB SUBSTATION to JONES CHAPEL 500kV. The project contributes 5.3% DFAX (5.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
GOLDEN GATE to EGE 100kV	Reconductor line	\$1,061,000	2% (\$26,000)
EGE to CUBA CITY 100kV	Reconductor line	\$2,397,000	2% (\$60,000)
	Total allocated to project		\$86,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.